Lab 6 - Activity 6 - Source Assurance, RAG Comparison & Maintenance

Day 2 | Topic 3 | 45 min | Multi-agent team + human oversight | LO3

Retrieve approved source passages, compare grounded and unsupported answers, and repair a stale source before it reaches the video.

Goal

Compare a retrieval-grounded assistant with a standard LLM and assess implementation and maintenance feasibility using actual source tests (LO3; PP Task 3 written analysis).

The business scenario

The first cut depends on price, date and promotion wording. Source quality must be checked before product claims are reused. Then transfer the design to a fictional financial-services advisory workflow.

You will deliver

- outputs/source/source-register.md - file IDs, versions, passages and owners
- outputs/source/retrieval-test.md and verifier-report.md - supported, missing and conflicting cases
- outputs/source/rag-feasibility.md - standard LLM comparison, costs, access, upkeep and evaluation

Who does what

- Agents: Distinct Hermes profiles: researcher, scriptwriter, producer, auditor and publisher
- You (the human): Own approval gates, inspect evidence and accept the exact version

Before you start

- Activity 5 real Hermes video team is working
- Same approved launch workspace and original source files

Concepts in action

- Separate agent state and output ownership
- Source provenance and independent verification
- Exact-version human decision and honest completion

Lab files (data/)

File	What it is
brand-guide.md	Brand voice, colours, platform formats
campaign-brief.md	Gula Melaka Cold Brew campaign brief (offer, KPIs, timeline)
compliance-rules.md	C1-C10 brand-safety and escalation rules
product-catalogue.csv	12 products and prices
video-qa-checklist.md	V1-V9 video QA checks
assets/brand-palette.png	Brand image asset
assets/cold-brew-product-card.png	Brand image asset
assets/kopi-kaki-logo.png	Brand image asset
source-test-cases.csv	

All company data in this lab (Kopi Kaki Co., people, reviews, comments, metrics, competitors) is fictional and generated for training.

Copy-ready prompts: `COPY-PROMPTS.txt` or the course Lab Prompts page. Printable instructions: `LAB-06-Instructions.pdf`.

Steps

1. Register sources (10 min)

Prompt - paste into *Hermes Desktop chat in the Kopi Kaki workspace*:

```
Assign the researcher a separate source-register.md task. Inspect campaign-brief.md, product-
catalogue.csv and compliance-rules.md. Record actual filenames, human-readable source IDs,
current version/date if present, owner and permitted use. Quote exact supporting passages for
price, launch date and promotion. Missing version metadata is a gap; do not invent it.
```

□ Expected: Actual passages and stable source IDs support video claims.

2. Run supported and unsupported retrieval tests (10 min)

Prompt - paste into *Hermes Desktop chat in the Kopi Kaki workspace*:

```
Use only the approved local sources for three tests: product price; full promo terms; whether
the drink has a certified health benefit. Quote the supporting passage and source ID for each
supported answer; answer INSUFFICIENT EVIDENCE for the unsupported health claim. Save
retrieval-test.md. This is a small file-retrieval grounding exercise, not proof of a deployed
vector database or production RAG system.
```

□ Expected: Supported answers cite real passages; unsupported claim is refused.

3. Verify independently and repair stale evidence (10 min)

Prompt - paste into *Hermes Desktop chat in the Kopi Kaki workspace*:

```
Assign the independent auditor to check each answer against the original source, not the
researcher summary. In a separate training test fixture copy one price line and change it to
an intentionally conflicting value; preserve the original files. Mark the fixture SYNTHETIC
STALE TEST, detect the conflict and keep downstream release gated until I choose the
authoritative source. Save verifier-report.md and the resolution; do not let majority vote
decide factual truth.
```

□ Expected: Verifier detects the seeded conflict and downstream copy uses the human-confirmed source.

4. Transfer to financial services and assess feasibility (15 min)

Prompt - paste into *Hermes Desktop chat in the Kopi Kaki workspace*:

```
Write rag-feasibility.md for a fictional financial-services client. Compare a standard LLM
assistant with retrieval from a firm-approved corpus: strengths, limitations, provenance,
latency, cost and access boundaries. Assess implementation feasibility: corpus owner,
ingestion/chunking/indexing choice, permissions, source versioning and rollback, refresh/re-
index routine, test questions, accuracy/relevance checks and human approval. Explain how this
local passage test differs from a production RAG service. Give one condition under which the
simpler standard assistant is adequate and one condition requiring retrieval.
```

□ Expected: Balanced RAG-vs-standard comparison with realistic maintenance plan and product decision.

Verification / checkpoint

[ ] Every accepted claim has a checked source passage

[ ] Auditor independently detects the stale test without changing approved originals

[ ] Feasibility covers cost/access/upkeep/evaluation and limitations of the local exercise

Checkpoint: Source-backed and missing-evidence tests plus independent verification and a written maintenance decision.

Troubleshooting

Symptom	Fix
Agent invents certification	Return to original passages; mark unsupported and block reuse.
Different source versions disagree	Human source owner decides; preserve old/new versions and retest affected artifacts.

Stretch challenge

Transfer the checks to a different product while preserving source and decision ownership.

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